

Exact weighted Pauli uncertainty on line graphs via the matching polytope

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Abstract

For Pauli observables, let the frustration graph join exactly the anticommuting pairs. We prove that the complete weighted quadratic uncertainty problem is exact on every line graph. For every finite simple graph R , the representation-independent beta body is exactly the matching polytope,

$$\text{BETA}(L(R)) = \text{MATCH}(R).$$

Equivalently, for every nonnegative edge weight w ,

$$\beta(L(R), w) = \alpha(L(R), w) = \nu(R, w),$$

where $\nu(R, w)$ is the maximum weight of a matching of R . The main proof maps the Hamiltonian to a Majorana bilinear. Nuclear-norm duality produces a skew contraction Q , and the squared supported entries Q_e^2 satisfy both the degree and all odd-set inequalities of Edmonds' matching polytope. It also yields the sharp matrix inequality

$$(\|A\|_*/2)^2 \leq \nu(R, w) \sum_{e \in E(R)} A_e^2/w_e$$

for every real skew matrix supported on R . The family $L(K_{2k+1})$, $k \geq 2$, proves that the theorem strictly extends the previously sufficient h -perfect route. An exhaustive small-root ablation separates the mechanism: degree constraints are exact on 66.89% of 6225 weighted instances, cliques plus odd cycles on 84.69%, and the complete odd-set system on 100%. Removing skewness fails already on a triangle. A dependency-free C# oracle independently checks 3808 random weighted skew directions on 119 roots with zero violations and 952 matching-supported equality directions to 7.78×10^{-16} . As a secondary extension, we prove that every simplicial claw-free quasi-line graph is rank-perfect and hence h -perfect. The analytic theorem, not the finite computation, carries the claim.

Status and scope. This is a self-contained manuscript that has not yet undergone external peer review. The proof uses classical matching-polytope and Majorana tools, which are cited explicitly. A documented search found no prior statement of the body identity or all-weight equality above, but a negative literature search is not a proof of priority. The expanded C045 search was frozen on 16 September 2026 and found no exact collision, but this remains support rather than certification. A C046 mock-referee pass formalized the body-level statement and audited every dependency in the central proof. No quantum advantage or new matching algorithm is claimed.

1 Introduction

Let $P_1, \dots, P_n$ be Hermitian Pauli observables. Their frustration graph G has vertex set $[n]$ and an edge ij exactly when $P_i P_j = -P_j P_i$; nonedges commute. For nonnegative w , define

$$\beta(G, w) = \max_{\rho} \sum_i w_i \text{tr}(\rho P_i)^2, \quad \alpha(G, w) = \max_{S \text{ stable in } G} \sum_{i \in S} w_i. \quad (1)$$

Here the first maximum may use any Pauli realization with exactly the prescribed pairwise relations: its value is representation-independent by the beta-number framework of Xu et al. [4, 5]. Since the observables in a stable set commute, a common eigenstate gives $\beta(G, w) \geq \alpha(G, w)$. The difficult direction is the upper bound.

The equality $\beta(G, w) = \alpha(G, w)$ for every nonnegative w is called $\hbar$ -perfection [4]. Known sufficient classes do not settle all line graphs. Line graphs are also the natural frustration graphs of quadratic Majorana Hamiltonians [2, 3], while their stable sets are precisely matchings. This suggests that a quantum uncertainty body and a classical matching polytope might coincide. The missing issue is not the degree constraints but the entire family of blossom, or odd-set, constraints.

Our contribution is a direct mechanism for all of them. If Q is a skew contraction, every odd principal submatrix loses at least one rank. This turns its Frobenius norm into the corresponding odd-set inequality. Together with row-norm degree constraints, Edmonds' theorem identifies the squared entries of Q as a point of the matching polytope. Weighted Cauchy–Schwarz then closes the quantum bound exactly.

The resulting theorem has three useful forms:

1. every finite line graph is $\hbar$ -perfect;
2. every nonnegative weighted instance is solved by maximum-weight matching, without state simulation;
3. a sharp support-sensitive nuclear-norm inequality holds for all real skew matrices.

The last form distinguishes the result from a merely unweighted skew-energy estimate. We also give an independent proof through known simplicial claw-free rank bounds and exhibit an infinite separation from the $\hbar$ -perfect sufficient class. Finally, rather than reporting only successful theorem checks, we compare the complete mechanism against two explicit weaker baselines and ablate skewness itself. As a secondary extension, we combine a simplicial-clique heredity lemma with the modern facet structure of fuzzy circular interval graphs to prove $\hbar$ -perfection for simplicial claw-free quasi-line graphs. This extension is separated from the headline contribution because it depends on the classical quasi-line facet classification.

2 Definitions and the variational reduction

For a fixed Pauli realization $\mathcal{S} = \{P_i\}_{i \in V(G)}$, write

$$\mathcal{Q}(\mathcal{S}) = \left\{ (\text{tr}(\rho P_i)^2)_{i \in V(G)} : \rho \text{ a state} \right\}.$$

If $T \subseteq \mathbb{R}^{V(G)}$, let $\downarrow T = \{x : \text{there is } y \in T \text{ with } 0 \leq x \leq y\}$. Proposition 1 of Xu et al. [4] proves that

$$\text{BETA}(G) = \text{conv}(\downarrow \mathcal{Q}(\mathcal{S})) \quad \text{and} \quad \beta(G, w) = \max_{y \in \text{BETA}(G)} w \cdot y \quad (w \geq 0) \quad (2)$$

are independent of the chosen realization. Thus $\text{BETA}(G)$ is a compact convex corner, whereas the raw joint range $\mathcal{Q}(\mathcal{S})$ need not be convex and can depend on the realization. This distinction is essential for the body-level statement below.

For a finite simple root graph $R = (V, E)$, its line graph $L(R)$ has one vertex for every edge of R ; two vertices are adjacent when their root edges share an endpoint. A matching of R is therefore a stable set of $L(R)$, and

$$\alpha(L(R), w) = \nu(R, w), \quad (3)$$

where $\nu(R, w)$ denotes maximum matching weight.

We use the following elementary variational identity. For fixed observables,

$$\max_{\rho} \sum_i w_i \operatorname{tr}(\rho P_i)^2 = \max_{\|a\|_2=1} \left\| \sum_i a_i \sqrt{w_i} P_i \right\|^2. \quad (4)$$

Indeed, Euclidean duality rewrites the squared norm of the expectation vector as a maximization over a . For fixed a , maximization over density matrices gives the squared spectral norm of the Hermitian linear combination. Compactness allows the two maxima to be interchanged. With $c_i = a_i \sqrt{w_i}$, the constraint is $\sum_i c_i^2 / w_i = 1$. Zero-weight coordinates may be deleted.

For a real matrix M , write $\|M\|_{\text{op}}$ for its operator norm, $\|M\|_F$ for its Frobenius norm, and $\|M\|_*$ for its nuclear norm. If $A = -A^\top$, then its nonzero singular values occur in equal pairs.

3 A skew-contraction lemma

Lemma 3.1 (Squared skew contractions lie in a matching polytope). *Let $R = (V, E)$ be finite and simple. If $Q = -Q^\top$ and $\|Q\|_{\text{op}} \leq 1$, define $x_e = Q_{uv}^2$ for every $e = uv \in E$. Then $x \in \text{MATCH}(R)$.*

Proof. Nonnegativity is immediate. For each $v \in V$,

$$\sum_{e \in \delta(v)} x_e \leq \sum_{u \in V} Q_{vu}^2 = \|e_v^\top Q\|_2^2 \leq 1.$$

Now let $S \subseteq V$ have odd cardinality. The principal matrix $Q[S]$ is odd-order and skew-symmetric, hence singular and of rank at most $|S| - 1$. Compression cannot increase operator norm, so every singular value of $Q[S]$ is at most one. Consequently

$$2 \sum_{e \in E(S)} x_e \leq \|Q[S]\|_F^2 \leq \operatorname{rank} Q[S] \leq |S| - 1.$$

Thus x satisfies nonnegativity, all degree inequalities, and all odd-set inequalities. Edmonds' matching-polytope theorem [1] states that these inequalities describe $\text{MATCH}(R)$ exactly. $\square$

The lemma is support-monotone: entries of Q outside E are irrelevant and only strengthen the row and Frobenius estimates. This matters because the nuclear-norm dual optimizer need not share the sparsity pattern of A .

Theorem 3.2 (Sharp weighted skew-matrix inequality). *Let $w_e > 0$, and let $A = -A^\top$ be real and supported on $E(R)$. With $A_e = A_{uv}$ in any fixed orientation,*

$$\left(\frac{\|A\|_*}{2} \right)^2 \leq \nu(R, w) \sum_{e \in E} \frac{A_e^2}{w_e}. \quad (5)$$

The constant is attained for every R, w .

Proof. By nuclear/operator-norm duality, there is a real Q with $\|Q\|_{\text{op}} \leq 1$ and $\langle A, Q \rangle_F = \|A\|_*$. Replacing Q by $(Q - Q^\top)/2$ preserves its pairing with skew A and does not increase its operator norm. Hence take Q skew. Skew symmetry gives

$$\frac{\|A\|_*}{2} = \sum_{e \in E} A_e Q_e.$$

By Lemma 3.1, $x_e = Q_e^2$ belongs to $\text{MATCH}(R)$, so $\sum_e w_e Q_e^2 \leq \nu(R, w)$. Weighted Cauchy–Schwarz yields

$$\left(\sum_e A_e Q_e \right)^2 \leq \left(\sum_e \frac{A_e^2}{w_e} \right) \left(\sum_e w_e Q_e^2 \right),$$

which proves (5). For sharpness, choose a maximum-weight matching M and set $A_e = w_e$ on M and zero elsewhere. Then $\|A\|_*/2 = \sum_{e \in M} w_e = \nu(R, w)$ and both sides equal $\nu(R, w)^2$. $\square$

4 Exact Pauli uncertainty on line graphs

Theorem 4.1 (Exact beta body of a line graph). *For every finite simple graph R ,*

$$\boxed{\text{BETA}(L(R)) = \text{MATCH}(R)}. \quad (6)$$

Consequently, for every nonnegative edge weight w ,

$$\boxed{\beta(L(R), w) = \alpha(L(R), w) = \nu(R, w)}. \quad (7)$$

Proof. Delete zero-weight edges. Assign a Majorana operator γ_v to each root vertex and represent $e = uv$ by $P_e = i\gamma_u\gamma_v$. After a Jordan–Wigner transform these are Pauli observables, and two anticommute exactly when the root edges share one endpoint [2]. By representation invariance [4], this realization computes $\beta(L(R), w)$.

For real c_e , define $A = -A^\top$ by $A_{uv} = c_e$ in a fixed orientation. Orthogonal skew normal form gives two-by-two blocks $\begin{pmatrix} 0 & s_j \\ -s_j & 0 \end{pmatrix}$. A compatible orthogonal change of Majorana generators turns $\sum_e c_e i\gamma_u\gamma_v$ into a sum of commuting involutions with coefficients s_j . Hence

$$\left\| \sum_e c_e i\gamma_u\gamma_v \right\| = \sum_j s_j = \frac{\|A\|_*}{2}. \quad (8)$$

If $|V(R)| = 2q$ or $2q + 1$, the skew normal form has q two-by-two blocks (and one zero direction in the odd case). The associated q commuting bilinear involutions have all joint sign patterns in the standard Clifford representation, which justifies the final equality in (8) also for odd-order roots.

Combining (4), Theorem 3.2, and the normalized constraint $\sum_e c_e^2/w_e = 1$ gives $\beta(L(R), w) \leq \nu(R, w)$.

Conversely, a maximum-weight matching is a stable set of $L(R)$. Its Pauli observables commute and have a common eigenstate, whose squared expectations on the matching are all one. Therefore $\beta(L(R), w) \geq \nu(R, w)$. Equation (3) proves the weighted identity for every $w \geq 0$. By (2), these are precisely the nonnegative support functions of $\text{BETA}(L(R))$; the same functions support $\text{MATCH}(R)$. Compact convex corners are determined by their nonnegative support functions, so (6) follows. $\square$

Corollary 4.2 (Realization-level blossom inequalities). *For any Pauli realization with frustration graph $L(R)$, every squared expectation profile $y_e = \text{tr}(\rho P_e)^2$ belongs to $\text{MATCH}(R)$. Conversely, every matching incidence vector belongs to $\text{BETA}(L(R))$.*

Proof. The first statement follows from $\mathcal{Q}(\mathcal{S}) \subseteq \text{BETA}(L(R)) = \text{MATCH}(R)$. For a matching, its commuting Pauli observables have a common eigenstate and hence squared expectation one on every matching edge. Downward closure then contains the matching incidence vector. $\square$

The theorem is constructive at the level of values: a maximum-weight matching algorithm returns both the exact quantum optimum and a commuting set attaining it. It does not assert an efficient circuit preparing a chosen common eigenstate under every hardware connectivity model.

5 An independent graph-polyhedral proof

Every line graph is claw-free. For a nonisolated root vertex v , the star $\delta(v)$ is a clique in $L(R)$, and for each edge $uv \in \delta(v)$ its neighbors outside the star are edges incident with u , hence form a clique. Thus each connected line graph admits a simplicial clique. Induced subgraphs of line graphs are again line graphs, so the class is hereditary in the form needed below.

The simplicial claw-free sector theorem of Chapman, Elman, and Mann [3] implies the rank inequality

$$\sum_{e \in U} \text{tr}(\rho P_e)^2 \leq \alpha(L(R)[U]) \quad (9)$$

for every induced support U . One obtains it by expressing the Hamiltonian norm through at most α nonnegative single-particle energies and applying Cauchy–Schwarz; this argument is representation-independent.

Apply (9) first to $U = \delta(v)$. Since this is a clique, $\sum_{e \in \delta(v)} y_e \leq 1$. For an odd root-vertex set S , apply it to $U = E(S)$. Then

$$\sum_{e \in E(S)} y_e \leq \alpha(L(R[S])) = \nu(R[S]) \leq \frac{|S| - 1}{2}.$$

Together with $y \geq 0$, these are exactly Edmonds’ inequalities, so $y \in \text{MATCH}(R)$. This reproves Theorem 4.1. The first proof is retained because it is direct and yields the stronger coefficient-sensitive matrix inequality (5).

6 Strictly beyond the h -perfect sufficient class

Xu et al. show that h -perfect graphs are $\bar{h}$ -perfect [4]. The converse route does not explain Theorem 4.1.

Theorem 6.1 (Infinite strict separation). *For every integer $k \geq 2$, $L(K_{2k+1})$ is $\bar{h}$ -perfect but not h -perfect.*

Proof. The first claim is Theorem 4.1. For the second, assign $x_e = 1/(2k)$ to every edge of K_{2k+1} . Maximal cliques of its line graph are root stars and triangles. Their values are respectively 1 and $3/(2k) \leq 1$. An induced odd cycle of length $\ell \geq 5$ has value $\ell/(2k) \leq (\ell - 1)/2$. Hence x satisfies nonnegativity, all clique inequalities, and all induced odd-cycle inequalities used by the h -perfect relaxation.

However,

$$\sum_e x_e = \frac{\binom{2k+1}{2}}{2k} = \frac{2k+1}{2}, \quad \nu(K_{2k+1}) = k.$$

The full-root odd-set inequality is violated by exactly $1/2$, so the stable-set polytope is strictly smaller than that relaxation. Thus the line graph is not h -perfect. $\square$

The witness shows why triangle and odd-hole checks alone miss the mechanism: the new proof supplies blossom inequalities for arbitrary odd vertex sets, not merely chordless cycles.

7 Secondary extension beyond line graphs: the rank-perfect SCF bridge

Call a graph *rank-perfect* when nonnegativity and the inequalities

$$\sum_{v \in U} x_v \leq \alpha(G[U]), \quad U \subseteq V(G), \quad (10)$$

describe $\text{STAB}(G)$. The SCF sector theorem used in Section 5 supplies (10) for every squared Pauli expectation profile whenever G is simplicial claw-free.

Theorem 7.1 (Rank-perfect SCF bridge). *Every rank-perfect simplicial claw-free graph is $\hbar$ -perfect.*

Proof. Let y be any squared expectation profile. The SCF rank theorem gives nonnegativity and (10); rank-perfectness therefore implies $y \in \text{STAB}(G)$. Hence $\beta(G, w) \leq \alpha(G, w)$ for every $w \geq 0$. The reverse inequality follows universally by choosing a common eigenstate for the commuting Pauli observables indexed by a stable set. Thus equality holds for every nonnegative weight. $\square$

Semi-line graphs are rank-perfect [6, 7], giving an immediate named class beyond the matching construction. The following argument closes the full quasi-line boundary under the SCF hypothesis.

Corollary 7.2. *Every simplicial claw-free semi-line graph is $\hbar$ -perfect.*

Lemma 7.3 (SCF heredity). *Every induced subgraph of a simplicial claw-free graph is componentwise simplicial claw-free.*

Proof. It is enough to delete one vertex z from a connected graph G with simplicial clique K . If $K \setminus \{z\}$ is nonempty, it remains a simplicial clique in its component of $G - z$. Consider any other component D . Connectivity gives a nonempty set $A = N_G(z) \cap V(D)$. This set is a clique: otherwise two nonadjacent vertices of A , together with a neighbor of z in the component containing $K \setminus \{z\}$, form a claw centered at z . If $K = \{z\}$, then $N_G(z)$ is already a clique, so the same conclusion holds in every component. Finally, for $u \in A$, two nonadjacent vertices of $N_D(u) \setminus A$ would form a claw with z centered at u . Thus A is a simplicial clique of D . Repeated deletion proves the claim. $\square$

Write $C(n, p)$ for the graph on $\mathbb{Z}_n$ in which two vertices are adjacent when their circular distance is at most $p - 1$.

Lemma 7.4 (Circulant obstruction). *If $n > 2p \geq 4$, then $C(n, p)$ has a simplicial clique if and only if $p = 2$.*

Proof. For $p = 2$ the graph is a cycle and each edge is a simplicial clique. Assume $p \geq 3$. First suppose that a clique K is *local*, meaning that it is contained in a block of p consecutive vertices. Rotate the block so that the first and last vertices of K are 0 and d , with $0 \leq d \leq p - 1$. If $d \leq p - 2$, then $x = -(p - d - 1)$ and $y = d + 1$ are outside K , both neighbor 0, and are nonadjacent because their circular separation is p . If $d = p - 1$ but the block has a missing vertex y , use $x = -(p - y)$ instead. The same separation argument applies. The only remaining case is the full block $K = \{0, \dots, p - 1\}$. Vertex 1 then has the two outside neighbors -1 and p ; their two circular separations are $p + 1$ and $n - p - 1$, both larger than $p - 1$. Thus no local clique is simplicial.

For completeness, we prove the required discrete circular-diameter bound directly. Put $r = p - 1$ and $q = n - r$. Rotate a clique S so that $0 \in S$, and write

$$A = \{a \in [1, r] : -a \in S\}, \quad B = \{b \in [1, r] : b \in S\}.$$

Thus $S = \{0\} \cup (-A) \cup B$. If either side is empty, or if $\max A + \max B \leq r$, then S is local. Otherwise put $a = \max A$, $b = \max B$. Adjacency of $-a$ and b forces $a + b \geq q$; write $h = a + b - q \geq 0$.

Compatibility with b forces every element of A to lie either in $[1, r-b]$ or in $[q-b, a] = [a-h, a]$. Symmetrically, every element of B lies either in $[1, r-a]$ or in $[b-h, b]$. For the two high tails define their deficit sets

$$X = \{a - x : x \in A \cap [a-h, a]\}, \quad Y = \{b - y : y \in B \cap [b-h, b]\}.$$

Both are subsets of $\{0, \dots, h\}$. We have $h+1 \notin X+Y$: equality would give a cross-distance $a+b-(h+1) = q-1$, strictly between r and q , so the corresponding two vertices would be nonadjacent. Consequently X and $h+1-Y = \{h+1-y : y \in Y\}$ are disjoint subsets of $\{0, \dots, h+1\}$, and hence $|X| + |Y| \leq h+2$. Therefore

$$\begin{aligned} |S| &= 1 + |A| + |B| \\ &\leq 1 + (r-b) + (r-a) + (h+2) \\ &= 3r - n + 3 = 3p - n. \end{aligned} \tag{11}$$

Apply this fact to a nonlocal clique K . The graph is $2(p-1)$ -regular and has clique number p . If K were simplicial, then for any $v \in K$ the set $N(v) \setminus K$ would be a clique, so $2p-1-|K| \leq p$ and hence $|K| \geq p-1$. When $n \geq 2p+2$, equation (11) instead gives $|K| \leq p-2$, a contradiction. At the remaining boundary $n = 2p+1$, $C(n, p)$ is the complement of the odd cycle C_{2p+1} . A clique K is therefore a stable set in that cycle. If $|K| \geq 3$, the cyclic gaps between successive vertices of K are at least two; since $|K| \leq p$, they sum to $2p+1 > 2|K|$, so some gap contains two consecutive internal vertices. Choose a vertex of K not incident with that gap. The two internal vertices both belong to $N(v) \setminus K$ in the antihole but are nonadjacent there, contradicting simpliciality. If $|K| = 1$, deleting v and its two cycle neighbors leaves a path containing a consecutive pair. If $K = \{u, v\}$, the longer $u-v$ arc has length at least four; the vertices at distances two and three from v along that arc give the same obstruction. Thus no boundary clique is simplicial. This also gives a direct proof of the odd-antihole observation recorded in [9]. $\square$

Theorem 7.5 (SCF quasi-line theorem). *Every simplicial claw-free quasi-line graph is rank-perfect and hence $\bar{h}$ -perfect.*

Proof. It suffices to consider a connected graph G . Quasi-line graphs outside the fuzzy circular interval class have only rank facets [7]. For a fuzzy circular interval graph, the facet theorem of Oriolo and Stauffer [8] says that every remaining facet is a clique-family inequality associated with a maximal (n, p) clique-circulant, where $\gcd(n, p) = 1$ and $n > 2p \geq 4$. By definition that clique-circulant contains an induced $C(n, p)$. Lemma 7.3 makes this induced graph SCF, so Lemma 7.4 forces $p = 2$. Coprimality then makes n odd. In the clique-family inequality, $r = n \bmod p = 1$, so the two coefficients $p-r$ and $p-r-1$ are respectively 1 and 0. The inequality is therefore a rank inequality, contradicting the assumption of a nonrank facet. Thus every nontrivial facet of $\text{STAB}(G)$ is rank, which is exactly rank-perfectness. The $\bar{h}$ -perfect conclusion follows from Theorem 7.1. $\square$

The extension is strict. The graph with graph6 code `H?`adQY` has nine vertices and twelve edges. Exact classification certifies that it is SCF and quasi-line but not a line graph. Its stable-set polytope has seventeen facets: nine nonnegativity facets and eight rank facets. One rank facet is supported on eight vertices, has stability number three, and its support is neither a clique nor an odd hole. The graph is therefore rank-perfect but not $\bar{h}$ -perfect, while Theorem 7.1 proves it $\bar{h}$ -perfect.

Figure 1 reports the adversarial finite boundary audit. Among all 4308 SCF graphs in the exact order-nine census, all 3758 quasi-line members have only rank facets; all 550 records with nonrank facets lie outside quasi-line. A further seeded stress contains 60 connected SCF, quasi-line, non-line proper circular-arc graphs at orders 10–12, with no discovered nonrank facet.

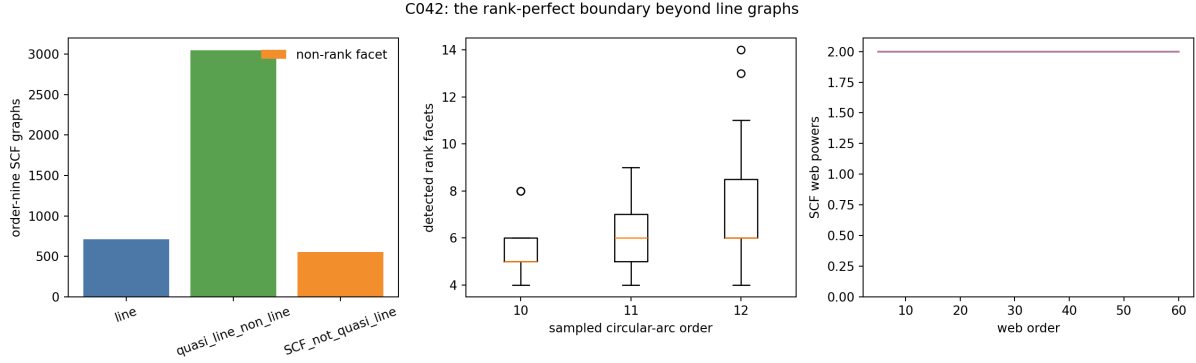


Figure 1: Rank-perfect boundary audit preceding the analytic closure. The order-nine panel is exhaustive for the frozen SCF census; the larger proper circular-arc panel is a conditioned seeded sample, and the web panel is an exact family sweep. The C043 proof no longer depends on these finite observations.

Candidate facets were found numerically and every reported rank facet was checked by exact validity and affine-rank calculations.

The SCF hypothesis is essential. The unrestricted statement that all quasi-line graphs are $\hbar$ -perfect is false: the anti-heptagon is quasi-line, has no simplicial clique, and is the smallest known $\hbar$ -imperfect graph [4]. The theorem likewise makes no claim for SCF graphs outside the quasi-line class.

8 Baseline comparison and mechanism ablations

The proof identifies a specific chain—skew contraction, squared entries, and the complete matching polytope. We tested whether any named link is dispensable. The benchmark contains every nonempty unlabeled root in the NetworkX graph atlas through order seven. Each of the 1245 roots receives one uniform and four seeded positive integer weight vectors, for 6225 weighted instances. This is exhaustive over the small unweighted root types, but the four nonuniform weights are stress cases rather than a probability model for applications.

For each instance we compare the exact maximum-weight matching against three linear relaxations on the same edge coordinates: degree inequalities only; degree, root-triangle clique, and all simple odd-root-cycle inequalities; and degree plus every odd-root-vertex-set inequality. In line-graph coordinates, the second is the clique-and-odd-hole system used by the $\hbar$ -perfect route, while the third is Edmonds’ full matching polytope.

Table 1: Relaxation value divided by exact maximum matching weight over 6225 weighted instances. “Exact” uses relative tolerance 2×10^{-7} .

Baseline	Mean	Median	95th pct.	Maximum	Exact
Degree only	1.0351	1.0000	1.1667	1.5000	66.89%
Cliques + odd cycles	1.0168	1.0000	1.1667	1.2500	84.69%
All odd sets	1.0000	1.0000	1.0000	1.0000	100.00%

Degree-only is strict on 2061 weighted cases from 797 roots; adding the clique-and-odd-cycle constraints still leaves 953 strict cases from 555 roots. The full odd-set system agrees with the discrete matching value in every instance. The maximum degree gap is the uniform triangle, $3/2$ versus 1. The maximum second-baseline gap is $5/2$ versus 2 on a five-vertex, nine-edge root; the same $5/4$ ratio occurs for K_5 . Figure 2 shows both the aggregate distribution and its dependence

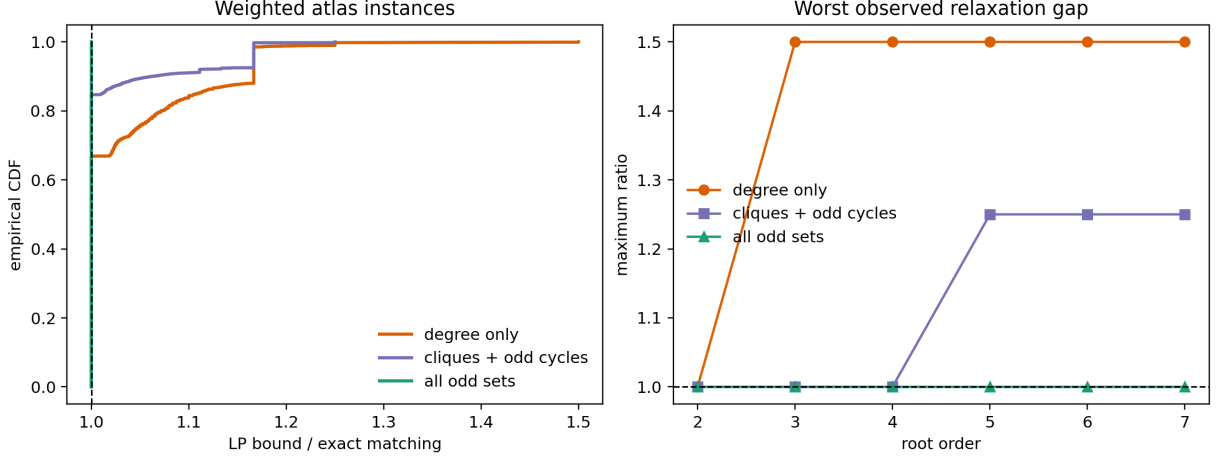


Figure 2: Weaker relaxations systematically overestimate weighted matching. The complete blossom system closes every tested gap. Percentages describe this finite benchmark, not a population of quantum models.

on root order.

The bipartite subset is a positive control: all three descriptions are exact on all 710 weighted cases. Among the 5515 nonbipartite cases, degree-only is exact on 62.63%, the clique-and-odd-cycle baseline on 82.72%, and the full polytope on 100%. Thus the implementation recovers the classical bipartite boundary before detecting the missing nonbipartite inequalities.

Controlled families separate the two kinds of odd constraint. With uniform weights on an odd cycle C_n , the odd-cycle constraint repairs the degree-only value $n/2$ to the matching value $(n-1)/2$. On an odd complete root K_n , however, both weaker relaxations retain value $n/2$, exceeding the matching value by $1/2$. We checked $n = 5, 7, 9, 11, 13$. The complete-root blossom, not another local cycle inequality, is the missing object.

Skewness is also necessary. On K_3 , let $Q = I - (2/3)J$. This symmetric Householder matrix has operator norm one and each off-diagonal square is $4/9$. Every degree sum is only $8/9$, but the triangle odd-set sum is $4/3 > 1$. A general contraction can therefore pass all degree tests and fail the matching polytope. In contrast, the skew cross-product matrix generated by $(1, 1, 1)/\sqrt{3}$ has norm one, edge squares $1/3$, and triangle sum exactly one. The odd-dimensional rank loss used in Lemma 3.1 is precisely what disappears in the ablation.

Finally, the sharp constant should not be confused with typical saturation. Across 4980 seeded Gaussian coefficient directions, the median theorem ratio is 0.3992, the 95th percentile is 0.7360, and the maximum is one. After excluding roots with fewer than three edges or matching value below two, the corresponding values are 0.3968, 0.7129, and 0.9997. By contrast, one matching-supported construction for every root gives ratio exactly one in all 1245 cases. The theorem is an exact worst-case support function, not a claim that generic Hamiltonians nearly saturate it. Figure 3 collects these controlled checks.

9 Contribution hierarchy, priority boundary, and conclusion

The paper has one headline result: for every finite root graph and every nonnegative weight, the complete Pauli beta body of its line graph is the classical matching polytope. Lemma 3.1 is the mechanism, and Theorem 3.2 is its support-sensitive matrix form. The second graph-polyhedral proof, the strict $L(K_{2k+1})$ separation, and the central ablations are robustness checks and consequences of that result.

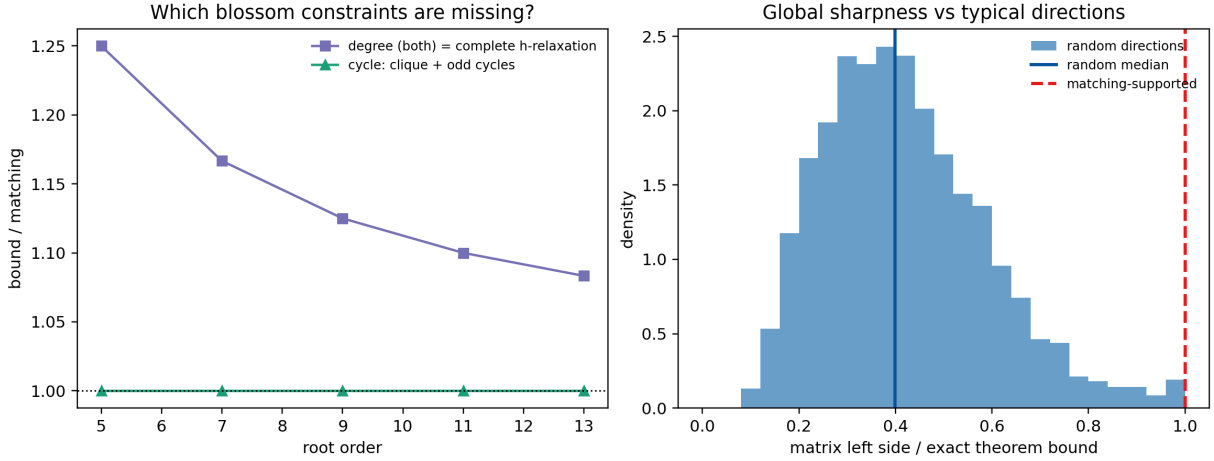


Figure 3: Mechanism ablations. Odd cycles repair cycle roots but not odd complete roots; removing skewness violates the blossom constraint; random directions are usually nonsaturating although equality is globally attainable.

The simplicial claw-free quasi-line theorem is a secondary structural extension. It establishes that the line-graph identity is not an isolated coincidence, but it relies on the published quasi-line facet classification and is not needed for Theorem 4.1. Keeping that dependency explicit also makes the main theorem independently assessable if a referee asks for the extension to be split off.

The closest prior literatures contain the ingredients separately. Beta numbers and representation invariance are due to Xu et al. [4, 5]; line-graph Majorana solvability is due to Chapman and Flammia [2]; and Edmonds supplies the matching polytope and weighted oracle [1]. Skew-energy and trace-norm work relates spectral quantities to matching, rank, or degree [12, 13], while McNulty studies a different line-graph invariant, measurement incompatibility [11]. The priority-sensitive conjunction here is the all-weight beta/matching identity, the generation of every blossom inequality by odd skew contractions, and the arbitrary-entry weighted matrix form. A source-by-source C045 audit found no exact statement of that conjunction. Negative search cannot establish priority, so we make no “first” claim and retain external refereeing as the independent check.

In summary, odd principal submatrices of a skew contraction lose one rank. That fact supplies every blossom inequality and converts an exact quantum uncertainty problem into maximum-weight matching. The result is analytic, arbitrary-size, and all-weight; it is not a hardware, speedup, or new matching-algorithm claim. The appendices report the broader computational falsification, proof-hardening ledger, and expanded source boundary.

A Scaling, density, and weak coupling

The atlas is exhaustive but small. A separate preregistered campaign samples 180 sparse roots at orders 10, 14, 18, 22, 26, 30: six replicates from each of five generators, with uniform, integer, lognormal, and Pareto edge weights. For each of 720 weighted instances we compare a nested hierarchy: degree constraints, all line-graph clique constraints, and every simple odd-root-cycle constraint through lengths five, seven, and nine. The reference value is exact maximum-weight matching. Forty cases at orders 10 and 14 also enumerate every odd vertex set; their full blossom LPs all equal matching.

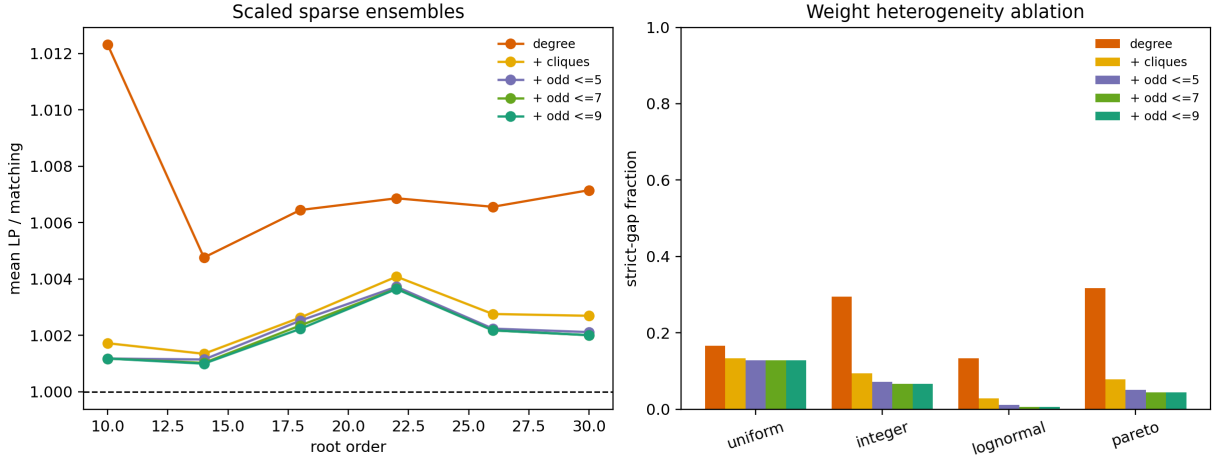


Figure 4: Scaling and weight ablations on 720 weighted sparse instances. Local constraints usually make the remaining gap small, but exactness depends on the graph and weight ensemble.

Table 2: Scaled sparse ensemble. Ratios are LP value divided by exact matching. The odd-cycle rows are truncated at the displayed length.

Baseline	Mean	95th pct.	Maximum	Exact
Degree only	1.007353	1.045550	1.500000	77.22%
All cliques	1.002541	1.013284	1.166667	91.67%
Odd cycles ≤ 5	1.002156	1.008641	1.166667	93.47%
Odd cycles ≤ 7	1.002068	1.005697	1.166667	93.89%
Odd cycles ≤ 9	1.002041	1.005697	1.166667	93.89%

Figure 4 shows that local constraints are numerically close on most sampled sparse graphs, especially with heterogeneous weights, but do not converge to universal exactness. Under the odd-through-nine baseline, exact fractions are 87.22%, 93.33%, 99.44%, and 95.56% for uniform, integer, lognormal, and Pareto weights. These values describe the frozen generators and seeds, not an application distribution.

Controlled direct sums prevent a misleading “typical equals sufficient” interpretation. Disjoint $K_3, C_5, C_7, C_9, C_{11}$ components retain ratios $3/2, 5/4, 7/6, 9/8, 11/10$ until the corresponding missing local constraint is added. More importantly, disjoint K_5 components retain ratio $5/4$ after all odd cycles through nine, although every simple cycle within a component is already covered. The missing inequality is the full five-vertex blossom. Orders reach 132, so the gap is not a vanishing finite-size effect.

For 540 Gaussian matrix directions on the scaled roots, the median theorem ratio is 0.3315, the 95th percentile is 0.4982, and the maximum is 0.7418; 180 matching-supported controls all attain one. Removing skewness also scales: block-diagonal copies of the symmetric Householder counterexample retain violation ratio $4/3$ through order 96. Figure 5 summarizes both controls.

A final density stress uses 45 Erdos–Renyi roots at orders 20, 30, and 40, expected degrees two through six, and two weight regimes. Odd-through-nine is strict in 5 of 90 weighted cases, with maximum ratio $29/28$. One graph already contains 384,484 enumerated odd cycles through length nine; the complete run takes 393.7 seconds. Thus explicit local-cycle enumeration can become large even when its remaining value gap is small.

To test robustness beyond disconnected unions, even numbers of K_5 blocks are joined in a ring by weak weighted bridges. With b blocks and bridge weight ε , the observed local optimum

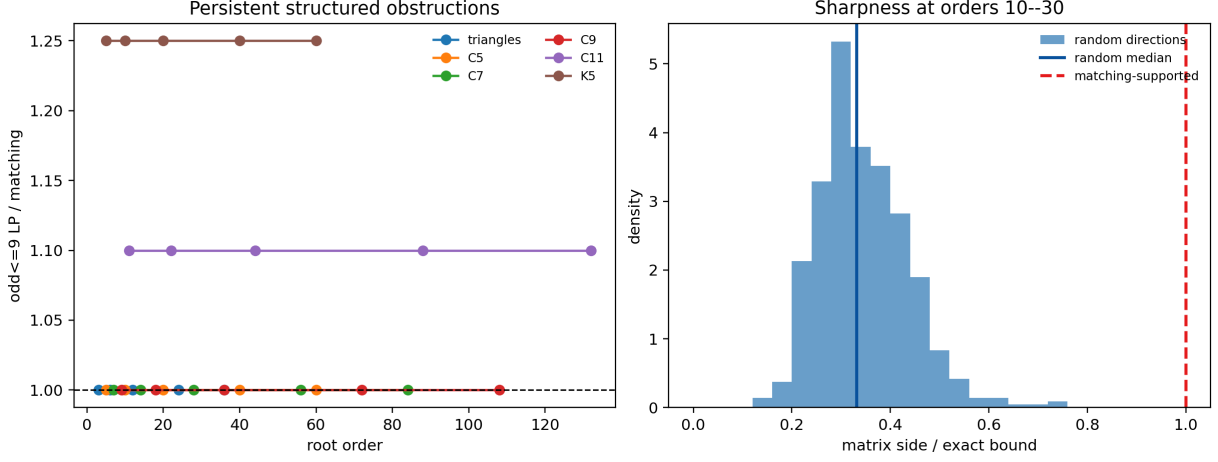


Figure 5: Left: local obstructions persist under direct sums. Right: generic Gaussian directions are far from equality while matching-supported directions remain exactly sharp.

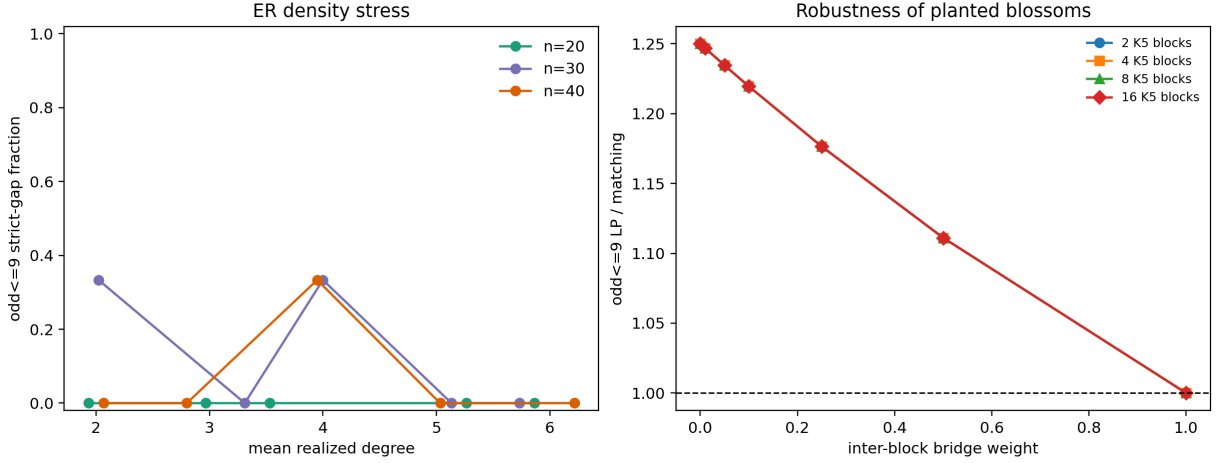


Figure 6: Density and weak-coupling stress. Random exactness is nonmonotone, whereas planted blossom gaps survive connectivity and weak bridges with a size-independent curve.

is $2.5b$ and matching is $2b + \varepsilon b/2$, giving ratio

$$\frac{2.5}{2 + \varepsilon/2}.$$

It is 1.2195 at $\varepsilon = 0.1$, 1.1111 at $\varepsilon = 0.5$, and closes only at unit bridge weight, identically for 2, 4, 8, and 16 blocks (up to 80 vertices). Figure 6 separates this planted mechanism from the nonmonotone random density sweep.

B Computational falsification and reproducibility

The proof is arbitrary-size and does not depend on enumeration. We nevertheless implemented a stress audit designed to expose normalization, orientation, and odd-set errors. It evaluates every nonempty graph in the NetworkX graph atlas, giving 1245 root graphs of order at most seven. For random positive weights and supported skew matrices, the code independently computes maximum-weight matching, the nuclear-norm expression, a polar dual optimizer, all degree and odd-set residuals, and the theorem ratio.

The largest observed ratio to the proved bound is 1.0000000000000002. The maximum

polar-pairing discrepancy is 3.20×10^{-14} ; maximum degree and odd-set excesses are 1.33×10^{-15} and 2.22×10^{-15} . In 47 small-root cases a dense Jordan–Wigner construction directly checks (8); its maximum error is 7.11×10^{-15} . Exact rational records for K_5, K_7, K_9 check the strict separation. A standard-library verifier checks the frozen summary and source hash, while five tests deliberately corrupt the gap, scope, hash, or norm data and require rejection.

The baseline-and-ablation run is independently frozen in a second JSON file, three CSV tables, and two figures. Its verifier recomputes every aggregate, checks all 6225 full-polytope equalities, proves the Householder counterexample with rational arithmetic, checks the controlled-family formulas, and verifies SHA-256 hashes of all five derived artifacts. Five additional corruption tests require rejection of altered ratios, counterexample entries, hashes, or scope flags.

The two scaled records add 225 sampled roots, 810 weighted random instances, 58 structured or coupling rows, ten hashed tables or figures, two independent standard-library verifiers, and eleven corruption tests. Their scope fields explicitly distinguish seeded ensembles, truncated odd-cycle systems, and controlled constructions from exhaustive or application-level claims.

The C042 record adds an exhaustive reclassification of 4308 order-nine SCF graphs, 868 web graphs, and 60 conditioned proper circular-arc stress graphs. Its independent verifier reconstructs the graph classes, strict witness, and exact rank-facet certificates. Six tests include five intentional corruptions.

The C043 record independently checks 4308 deterministic induced deletions, 6162 circulant parameter pairs through order 160, all 18,149 cliques in the small circulants through order 18, one exact published nonrank-web positive control, and 250 exact clique-family stress graphs. The small exact sweep explicitly includes the nonlocal clique $\{0, 3, 6\}$ in $C(9, 4)$ that falsifies the tempting local-cliques-only proof. Its standard-library verifier reconstructs the circulant witnesses and the positive-control facet; five mutation tests must reject corrupted records.

C044 removes the only geometric proof sketch from the new theorem. Its left/right sumset argument is fully contained in Lemma 7.4. The corresponding audit checks all 38,772 one-vertex deletions of the frozen SCF census, 1,027,351 admissible algebra rows through order 160, 4,194,302 deficit subsets, and all 3,803,174 cliques of every $C(n, p)$ through order 30. No simplicial clique occurs for $p \geq 3$. A separate standard-library verifier recomputes the algebra and sumset counts, checks all frozen hashes, and is protected by five mutation tests.

C046 is a claim-level mock-referee audit of the headline proof. It separates the realization-dependent raw joint range from the invariant beta body, checks the compact-convex support-function step, and makes the odd-order Clifford normalization explicit. Its verifier rejects a missing body identity, the old representation quantifier, an altered report, and false review-status flags. This is an internal adversarial check, not external peer review.

C047 adds a genuinely independent runtime check of the sharp weighted skew-matrix inequality. A C# program compiled by the standard .NET Framework compiler uses no Python, NumPy, SciPy, NetworkX, or external linear-algebra package. It implements subset-dynamic-programming maximum matching and a Jacobi eigensolver for $-A^2$ from scratch. On 119 structured and seeded roots, 952 positive weight vectors, and 3808 Gaussian skew directions, it finds zero violations; the largest ratio is 1.0000000000000002. A further 952 matching-supported directions attain equality with maximum residual 7.78×10^{-16} . This is cross-language falsification of the analytic inequality, not formal verification or external peer review.

From the repository root:

```
python experiments/pauli_fourth_moment_phase0/
run_c038_line_graph_hbar.py

python -S experiments/pauli_fourth_moment_phase0/
verify_c038_line_graph_hbar.py

python -m unittest discover -s
```

```

experiments/pauli_fourth_moment_phase0
-p test_c038_line_graph_hbar.py -v

python experiments/pauli_fourth_moment_phase0/
run_c039_central_ablations.py -overwrite

python -S experiments/pauli_fourth_moment_phase0/
verify_c039_central_ablations.py

python -m unittest discover -s
experiments/pauli_fourth_moment_phase0
-p test_c039_central_ablations.py -v

python experiments/pauli_fourth_moment_phase0/
run_c040_scaled_ablations.py -overwrite

python -S experiments/pauli_fourth_moment_phase0/
verify_c040_scaled_ablations.py

python experiments/pauli_fourth_moment_phase0/
run_c041_density_coupling_stress.py -overwrite

python -S experiments/pauli_fourth_moment_phase0/
verify_c041_density_coupling_stress.py

python experiments/pauli_fourth_moment_phase0/
run_c042_rank_perfect_boundary.py

python -S experiments/pauli_fourth_moment_phase0/
verify_c042_rank_perfect_boundary.py

python experiments/pauli_fourth_moment_phase0/
run_c043_quasiline_scf_falsification.py -overwrite

python -S experiments/pauli_fourth_moment_phase0/
verify_c043_quasiline_scf_theorem.py

python experiments/pauli_fourth_moment_phase0/
run_c044_proof_hardening.py -overwrite

python -S experiments/pauli_fourth_moment_phase0/
verify_c044_proof_hardening.py

python -S experiments/pauli_fourth_moment_phase0/
verify_c046_mock_referee_body_audit.py

```

The wrapped paths are single commands. Numerical agreement is falsification support, not the proof of Theorem 4.1.

C Expanded relation to prior work and claim boundary

Chapman and Flammia identify line-graph frustration structures as the free fermion setting for quadratic Majorana Hamiltonians [2]. Chapman, Elman, and Mann develop the wider simplicial claw-free framework [3]. Xu et al. define weighted beta numbers and $\hbar$ -perfection and establish other sufficient graph classes [4]; their accessible manuscript does not state the all-line-graph weighted matching equality used here. Oriolo and Eisenbrand et al. supply the classical semi-line and quasi-line polyhedral boundary [6, 7]. Oriolo and Stauffer’s later clique-circulant facet theorem [8] is the structural input to Theorem 7.5.

The graphs $L(K_{2k+1})$ already appear in magic-state simulation via Jordan–Wigner transformations [10]; the separating graphs themselves are not new. McNulty studies measurement incompatibility on line graphs using skew-energy bounds [11]. Classical skew-energy inequalities

related to matching are studied by Tian and Wong [12]. The uniform shadow of (5) is therefore close to known rank and skew-energy bounds. Our priority-sensitive claim is the exact all-weight Pauli beta/matching identity, its complete matching-polytope mechanism, and the sharp weighted matrix form. The underlying ingredients – Edmonds’ theorem, nuclear duality, Majorana normal form, and line-graph solvability – are not claimed new.

The repository contains a source-by-source adversarial audit with explicit overlap and falsification criteria. This supports careful positioning but cannot exclude differently worded or inaccessible prior results. Journal referees remain the independent priority check when no pre-submission expert is available.

D Extended discussion and scope

Odd principal submatrices of a skew contraction lose one rank. That elementary fact generates every blossom inequality and turns the quantum optimization on a line graph into the classical matching polytope. Consequently all finite line graphs are $\hbar$ -perfect for arbitrary nonnegative weights, and the associated skew-matrix inequality is sharp. The infinite family $L(K_{2k+1})$ shows that this mechanism is genuinely broader than the known h -perfect sufficient route.

The ablations sharpen that interpretation. Degree constraints alone explain the entire bipartite control but fail broadly outside it. Adding every clique and odd-cycle inequality reduces rather than eliminates the gap, while full odd-set constraints eliminate it. Removing skewness breaks the key conclusion on the smallest possible odd root. These are finite tests of the proof’s named ingredients, not empirical substitutes for the theorem.

Scaling clarifies the practical boundary. Local constraints are often nearly exact on seeded sparse ensembles, but constant gaps persist on direct sums and under weak coupling. Explicitly enumerating short odd cycles can require hundreds of thousands of rows by order 40, while the complete mathematical object remains maximum-weight matching rather than an ever longer list of local repairs.

The rank-perfect SCF bridge and the clique-circulant obstruction show that the line-graph mechanism is not the only analytic route. Exact Pauli uncertainty extends to every SCF quasi-line graph, including a certified graph that is neither line nor h -perfect. The unrestricted SCF and unrestricted quasi-line statements remain outside the theorem.

The result concerns exact uncertainty values, not quantum advantage or hardware performance. The next scientific tests are independent refereeing and a wider bibliographic priority search; adding more random instances would not strengthen the analytic theorem.

Data and code availability

Source, frozen records, verifier, and tests are available in the [public C038 GitHub branch](#). The repository index identifies the theorem note, priority audit, generator, frozen JSON, independent verifiers, and corruption-controlled tests. Every reported numerical constant is read from ten frozen JSON records by the manuscript checker. They include graph identifiers, generator seeds, weight regimes, relaxation values, aggregate summaries, explicit scope flags, and hashes for all derived tables and figures.

Declarations

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AI assistance. OpenAI Codex was used for code generation, computational verification, manuscript editing, and language polishing. The author reviewed all outputs and accepts full responsibility for the scientific content.

Author contributions. The author is solely responsible for the conception, theory, experiments, analysis, and writing of the manuscript. No QPU data are used in the theorem or its verification.

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